

AI Model Architectures: A Technical Compendium

A comprehensive reference document detailing major foundation models, neural network blueprints, and modern paradigms like BERT, GPT, and Diffusion.

Introduction

Artificial Intelligence models are structured on varying neural network frameworks optimized for specific sequence processing, perception, or generation tasks. This document breaks down major paradigms across Language, Vision, Generative, and Agentic AI.

1. Transformer Architectures (The Modern Standard)

Introduced in 2017, the Transformer architecture utilizes *self-attention mechanisms* to process data sequences in parallel rather than sequentially, replacing traditional recurrence models.

Encoder-Only (e.g., BERT)

BERT (Bidirectional Encoder Representations from Transformers) reads text from left to right and right to left simultaneously. It utilizes *Masked Language Modeling (MLM)* to predict hidden words in a sentence, making it highly optimal for understanding context, semantic search, classification, and named entity recognition (NER).

Decoder-Only (e.g., GPT, Llama)

Models like **GPT (Generative Pre-trained Transformer)** and Meta's Llama are *autoregressive*. They process text sequentially to predict the next word (causal language modeling), making them the backbone for fluid text generation and modern chat agents.

Encoder-Decoder (e.g., T5, BART)

These models use an explicit encoder to capture input text context and pass it to a separate decoder. This dual structure excels at sequence-to-sequence tasks like language translation, text summarization, and abstract code generation.

2. Vision & Multimodal Architectures

Convolutional Neural Networks (CNNs)

Classic vision models (e.g., ResNet) use localized grids or filters to extract spatial hierarchies from pixels. They remain critical for edge processing, medical imaging, and standard object detection.

Vision Transformers (ViTs)

ViTs break an image up into a sequence of flattened patches and pass them through a standard Transformer block. They scale exceptionally well with massive datasets and outperform CNNs on complex global context understanding.

Vision-Language Models (VLMs)

Models like CLIP cross-reference text embeddings with image embeddings in a shared mathematical space. This allows models to seamlessly read diagrams, interpret charts, and visually reason about complex photos.

3. Advanced Scaling & Generative Paradigms

Mixture of Experts (MoE)

Instead of activating the entire network for a basic prompt, MoE models route input tokens dynamically to smaller, specialized sub-networks ('experts'). This drastically improves compute efficiency and scales parameter counts (like Mixtral or GPT-4) without inflating runtime hardware cost.

Diffusion & Flow Matching Models

The standard for image generation (e.g., Stable Diffusion, Midjourney). These systems learn to iteratively remove Gaussian noise from a blank frame until a clear visual emerges, guided carefully by prompt embeddings.

Large Action Models (LAMs) & Agents

An evolutionary leap from static answering to autonomous planning. LAMs map high-level user intent into real-time sequence actions across computer operating systems, interacting directly with browser APIs and application buttons.

Architecture Blueprint Quick Summary

Type	Key Mechanisms	Primary Use Cases
Encoder-Only (BERT)	Bidirectional self-attention, MLM	Classification, Search, Sentiment
Decoder-Only (GPT)	Autoregressive, Causal attention	Text generation, Chat, Coding
Enc-Dec (T5)	Cross-attention block routing	Translation, Summarization
Mixture of Experts	Router gates, Active sub-networks	Compute-efficient ultra-scaling
Diffusion	Denoising score matching	Image/Video/Audio generation